Problem Statement

Water pollution has become one of the major environmental challenges in today's world. Checking water quality through lab tests is often slow, costly, and needs manual sampling. With the rise of technology, it's possible to make this process smarter and faster using AI.

This project focuses on building a **Convolutional Neural Network (CNN)** model that can automatically classify water images into three types — **Clean**, **Muddy**, and **Polluted** — by analyzing their visual patterns.

The main idea is to develop a simple and efficient image-based system that can help detect water quality directly from pictures captured by cameras or drones. This approach supports the goal of **environmental sustainability**, making it easier to monitor water health and pollution levels using modern deep learning techniques.

Objective

The main goal of this project is to create a deep learning model that can automatically identify the quality of water from its image — whether it's **Clean**, **Muddy**, or **Polluted**.

Through this project, I aim to:

- Use **Convolutional Neural Networks (CNN)** to perform image classification.
- Analyze the visual features of water such as color and clarity to predict its quality.
- Promote **environmental sustainability** by making water quality detection faster and easier through image-based analysis.
- Build a simple model that can later be connected with IoT devices, drones, or mobile applications for real-time monitoring.

Expected Outcome

- A working **CNN model** that can correctly classify water images into clean, muddy, and polluted categories.
- A well-organized dataset of labeled water images used for training and testing.
- Graphs showing the model's training and validation accuracy and loss.
- A small yet meaningful step toward using **AI for environmental health and pollution control**.

- (Optional) A basic **Flask or Streamlit web app** where users can upload an image and instantly check the predicted water quality.

Dataset Description

- The dataset used in this project contains images of different water conditions such as **Clean**, **Muddy**, and **Polluted** water bodies. Each image visually represents the quality of water based on color, clarity, and the presence of visible impurities.
- The dataset is divided into three folders — one for each category — which helps the CNN model learn to differentiate between them. The images were collected from open datasets and reliable sources like **Kaggle** and **Google Images** to ensure variety and balance among classes.
- This dataset is suitable for training deep learning models for **environmental monitoring and sustainability-based applications**.

Dataset Source

- **Primary dataset**:
  Clean Dirty Water Dataset – Kaggle
- **Alternative dataset (for additional samples)**:
  Water Pollution Images – Kaggle
- **Google Images (Manually Collected):**
  To improve the model's ability to recognize varied water conditions, additional images — especially for the muddy water category — were manually collected from Google Images.